

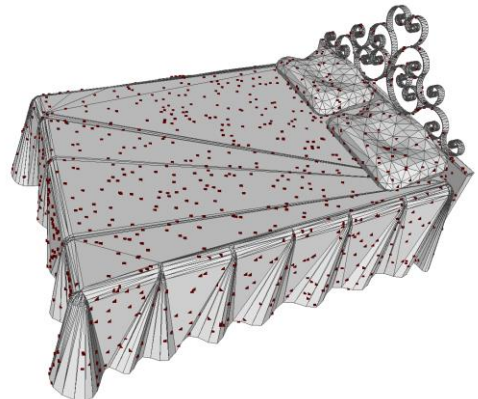
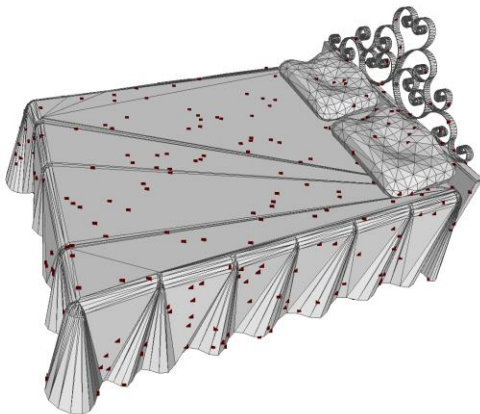
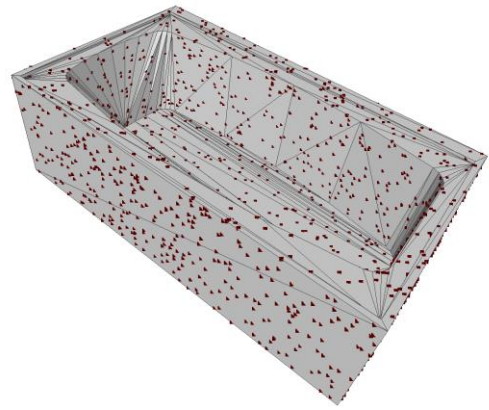
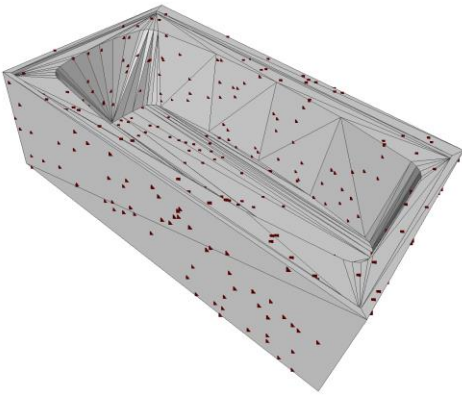
Assignment 4 Report

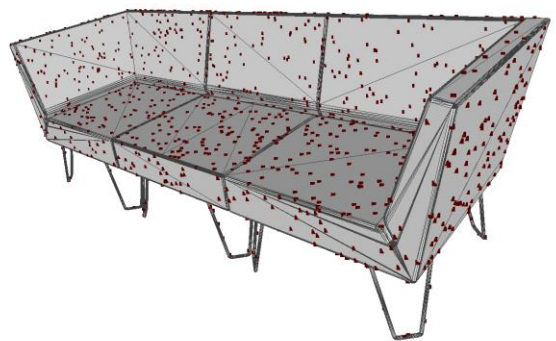
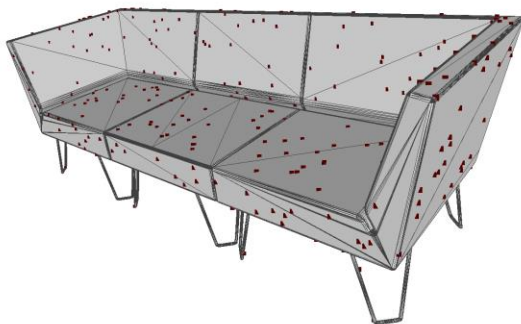
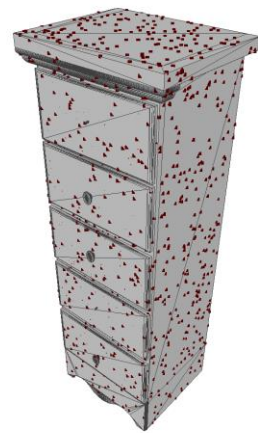
Sibei Zhou (301416917)

Part 1: Dataloader

Images of 5 meshes with their sampled point clouds at two different densities

(Left: 512; Right: 2024)





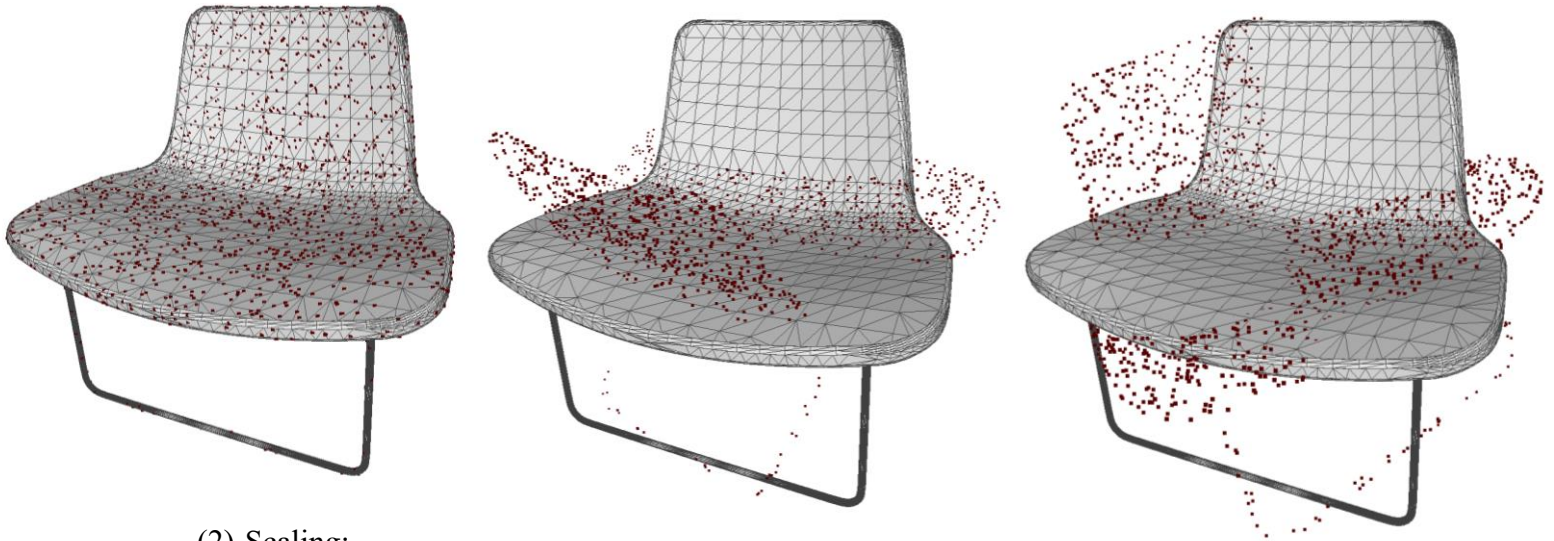
Part 2: Augmentation

For each augmentation submit an image of the mesh, originally sampled point cloud, and two variations generated using that augmentation. In the end, include one final example with all augmentations applied at once.

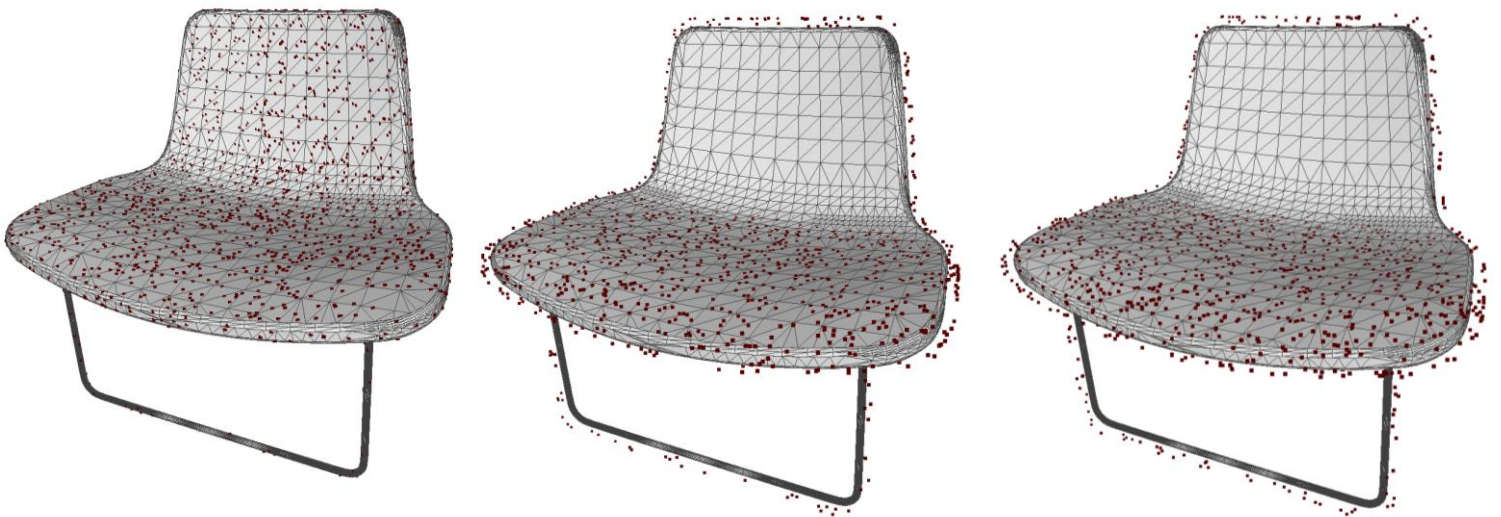
(Left: originally sampled point cloud;

Middle and Right: two variations generated using that augmentation)

(1) Rotation:



(2) Scaling:



(3) Translate:



(4) Jitter:

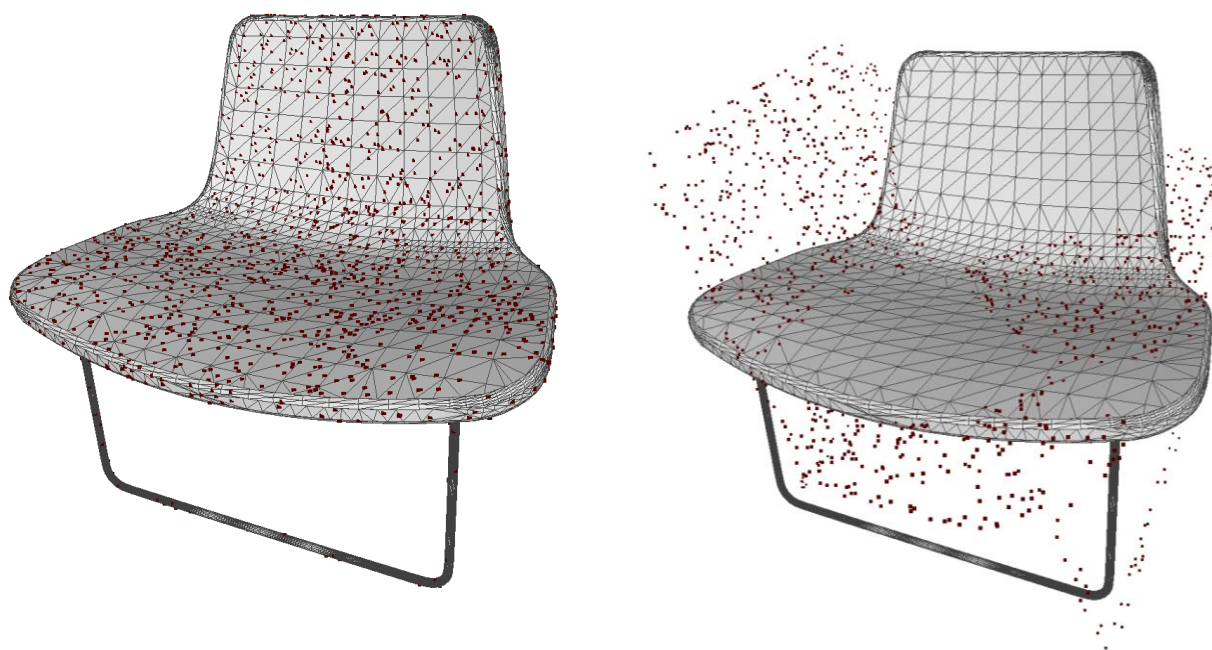


(5) Dropout:

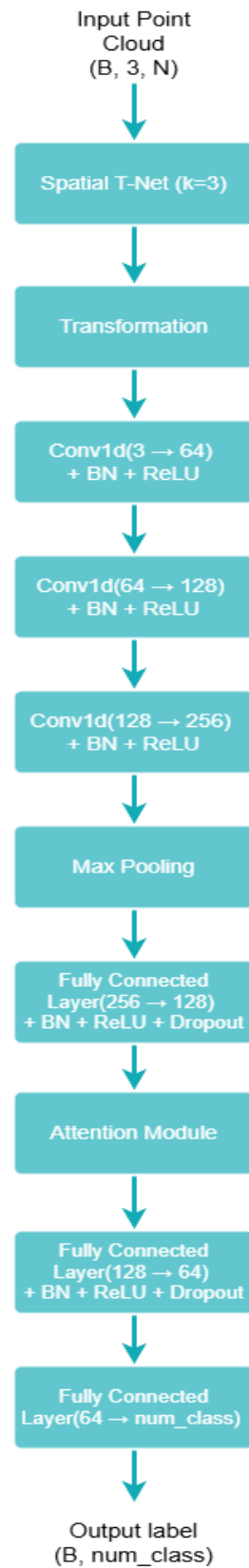


All augmentations applied:

(Left: originally sampled point cloud; Right: using all augmentations)



Part 3 Model Architecture:



Short Explanation of the Unique Model Design and Its Differences from PointNet.

(1) Attention Module for Feature Refinement:

After extracting the global feature vector via shared MLPs and max pooling, my model passes the features through a simple attention module. This module computes adaptive weights for each feature channel, allowing the network to emphasize more discriminative features. It might be useful for potentially improve classification performance by focusing on the most informative parts of the global representation.

(2) Simplified Feature Transform:

Unlike the original PointNet, which employs two transformation networks, my model only uses an input transformation while the feature transformation TNet has been removed. This simplification is based on the clean and well-aligned dataset, it reduces the model's complexity and computational overhead.

(3) Dropout layer:

The model integrates dropout layers before the final fully connected layers to mitigate overfitting.

Influential Work:

Qi, C. R., Su, H., Mo, K., & Guibas, L. J. (2017). PointNet: Deep Learning on Point Sets for 3D Classification and Segmentation. In Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR). <https://arxiv.org/abs/1612.00593>

Qi, C. R., Yi, L., Su, H., & Guibas, L. J. (2017). PointNet++: Deep Hierarchical Feature Learning on Point Sets in a Metric Space. In Advances in Neural Information Processing Systems (NeurIPS). <https://arxiv.org/abs/1706.02413>

Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, Ł., & Polosukhin, I. (2017). Attention Is All You Need. In Advances in Neural Information Processing Systems (NeurIPS). <https://arxiv.org/abs/1706.03762>

Part 4 Additional Questions:

1. Accuracy Metrics

Per-class accuracy:

Class 0 (bathtub): 76.00%

Class 1 (bed): 88.00%

Class 2 (chair): 99.00%

Class 3 (desk): 62.79%

Class 4 (dresser): 86.05%

Class 5 (monitor): 89.00%

Class 6 (night_stand): 76.74%

Class 7 (sofa): 93.00%

Class 8 (table): 94.00%

Class 9 (toilet): 88.00%

Overall accuracy: 86.23%

2. Visualization

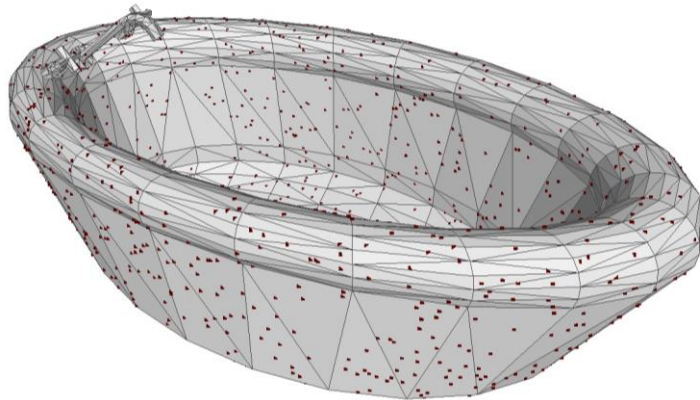
Images of 10 test meshes (one from each category), their sampled point clouds, and the top three predicted categories with probability scores.

Sample from Class bathtub

Top 1 prediction: bathtub (97.45%)

Top 2 prediction: bed (2.14%)

Top 3 prediction: sofa (0.23%)

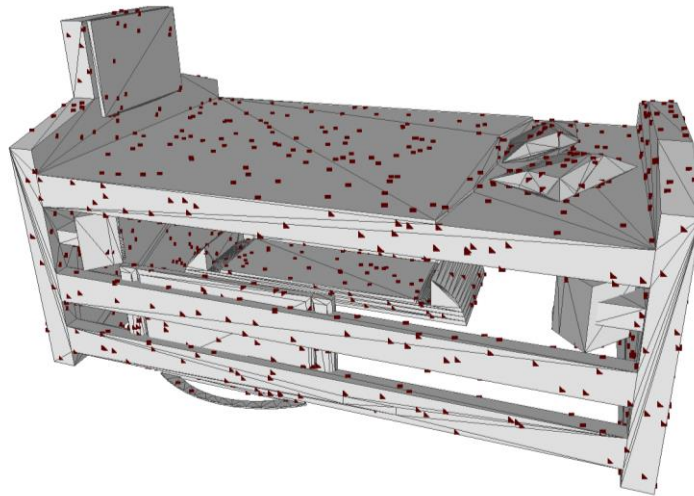


Sample from Class bed

Top 1 prediction: bed (78.30%)

Top 2 prediction: chair (7.01%)

Top 3 prediction: sofa (4.93%)

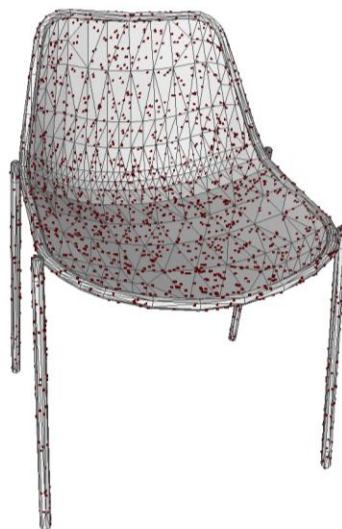


Sample from Class chair

Top 1 prediction: chair (99.97%)

Top 2 prediction: toilet (0.01%)

Top 3 prediction: night_stand (0.01%)



Sample from Class desk

Top 1 prediction: desk (40.83%)

Top 2 prediction: dresser (35.57%)

Top 3 prediction: toilet (7.74%)



Sample from Class dresser

Top 1 prediction: dresser (85.21%)

Top 2 prediction: night_stand (13.60%)

Top 3 prediction: monitor (0.56%)

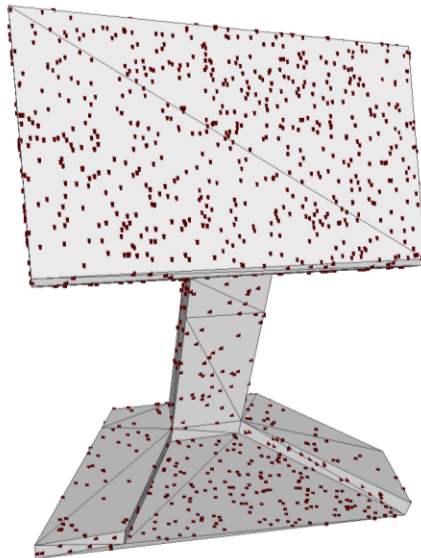


Sample from Class monitor

Top 1 prediction: monitor (71.74%)

Top 2 prediction: chair (12.30%)

Top 3 prediction: toilet (10.07%)

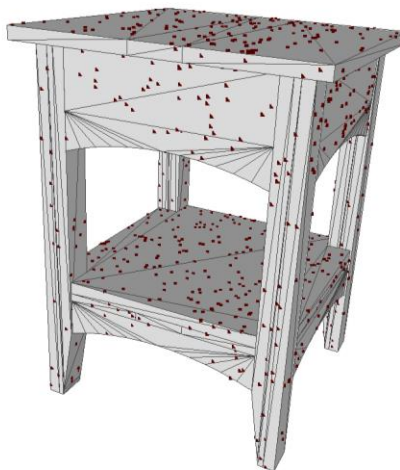


Sample from Class night_stand

Top 1 prediction: night_stand (93.72%)

Top 2 prediction: dresser (2.25%)

Top 3 prediction: table (1.93%)

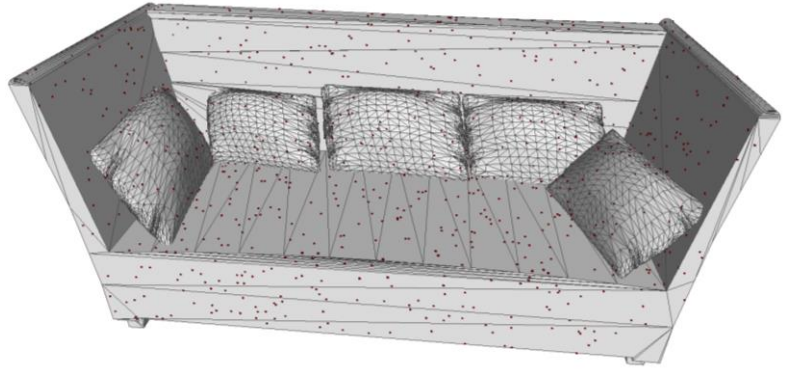


Sample from Class sofa

Top 1 prediction: sofa (97.82%)

Top 2 prediction: desk (1.21%)

Top 3 prediction: bed (0.46%)

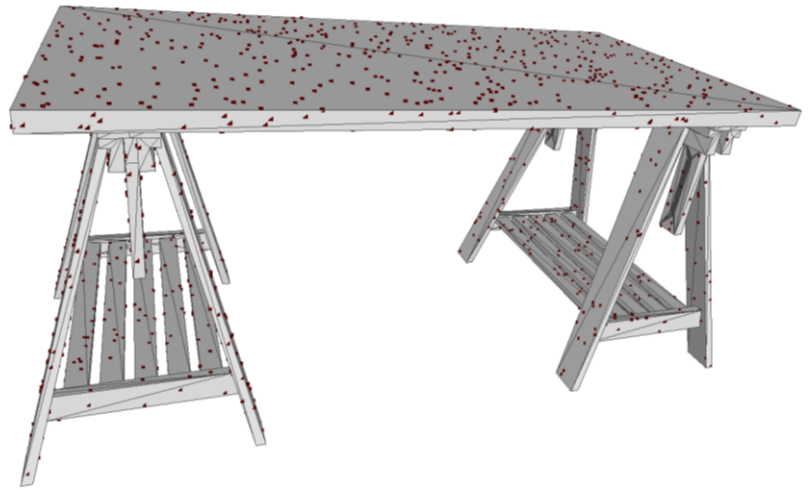


Sample from Class table

Top 1 prediction: table (88.24%)

Top 2 prediction: bed (6.14%)

Top 3 prediction: chair (3.19%)

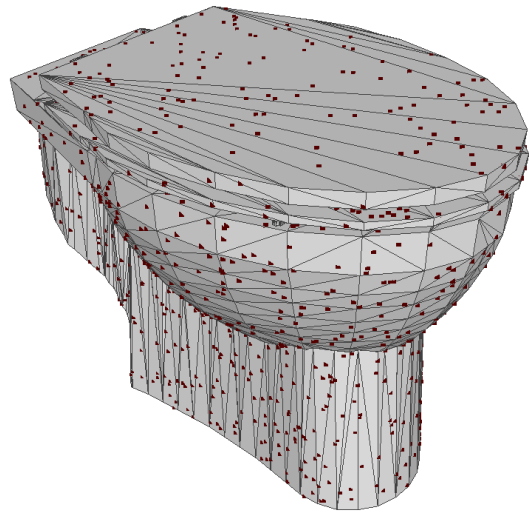


Sample from Class toilet

Top 1 prediction: toilet (90.37%)

Top 2 prediction: monitor (8.47%)

Top 3 prediction: night_stand (0.72%)



3. Model Strengths

(1) Chair (99.00%)

Why it's effective: Chairs generally have clear, consistent geometric structures—such as well-defined backrests and legs—which help the network learn distinct local and global features.

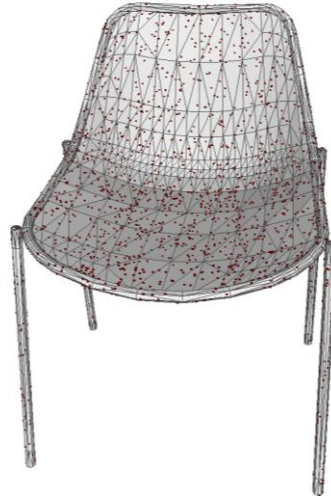
Example for chair:

Sample from Class chair

Top 1 prediction: chair (99.97%)

Top 2 prediction: toilet (0.01%)

Top 3 prediction: night_stand (0.01%)



(2) Table (94.00%) and Sofa (93.00%)

Why they are effective: Both tables and sofas tend to exhibit relatively stable, prototypical shapes like flat surfaces, rectangular forms, or distinctive curves that the network can recognize reliably.

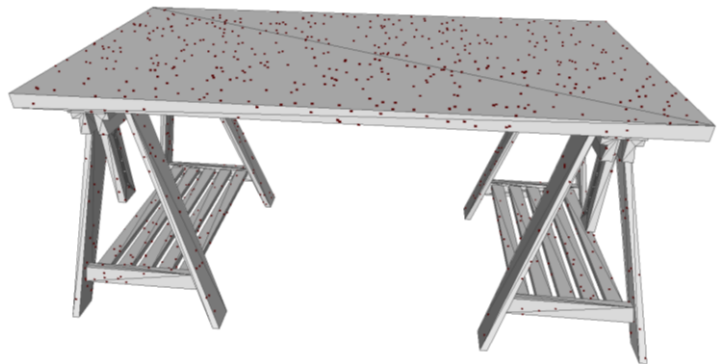
Example for table:

Sample from Class table

Top 1 prediction: table (94.29%)

Top 2 prediction: bed (3.08%)

Top 3 prediction: chair (1.06%)

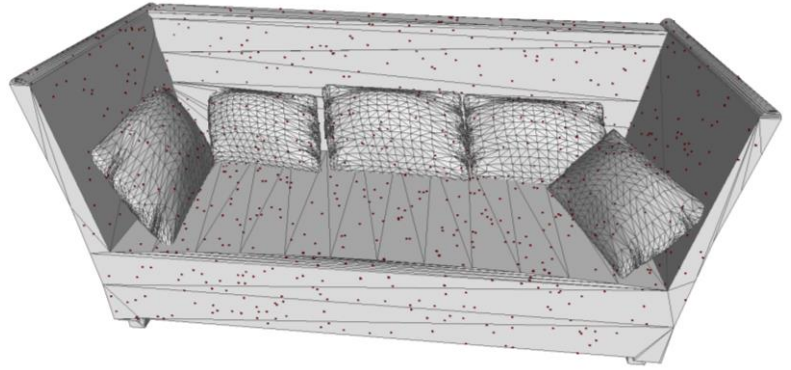


Sample from Class sofa

Top 1 prediction: sofa (97.82%)

Top 2 prediction: desk (1.21%)

Top 3 prediction: bed (0.46%)



4. Model Weaknesses

(1) Desk (62.79%)

Ambiguous Geometry: Desks might share similar features with other classes, such as tables, dressers or bed, especially if the point cloud does not capture fine details.

Intra-class Variability: Desks come in many forms (e.g., office desks vs. study desks) which can lead to increased variability in the point cloud samples.

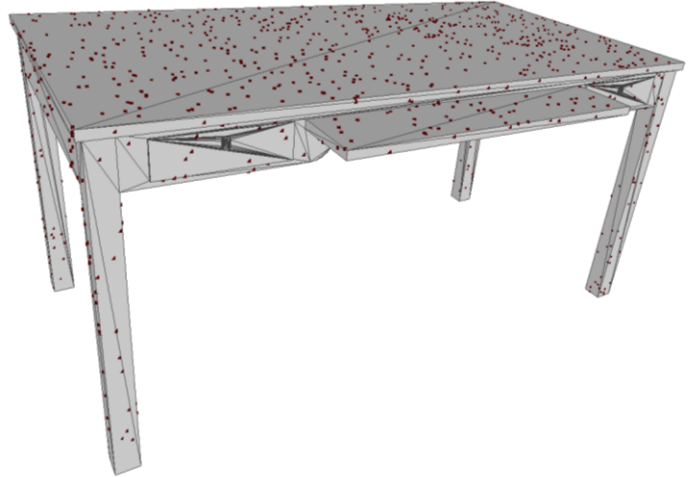
Example for desk:

Sample from Class desk (misclassified)

Top 1 prediction: table (95.71%)

Top 2 prediction: desk (4.24%)

Top 3 prediction: bed (0.03%)



(2) Bathtub (76.00%) and Night_stand (76.74%)

Complex or Varying Shapes: Bathtubs and night stands often have more subtle or complex curvature that may be affected by viewpoint changes or sampling inconsistencies.

In addition, in simplified point cloud representations, it might suffer from overlapping feature representations. Bathtub, which can sometimes appear similar to bed. And nightstands, which can sometimes appear similar to dressers.

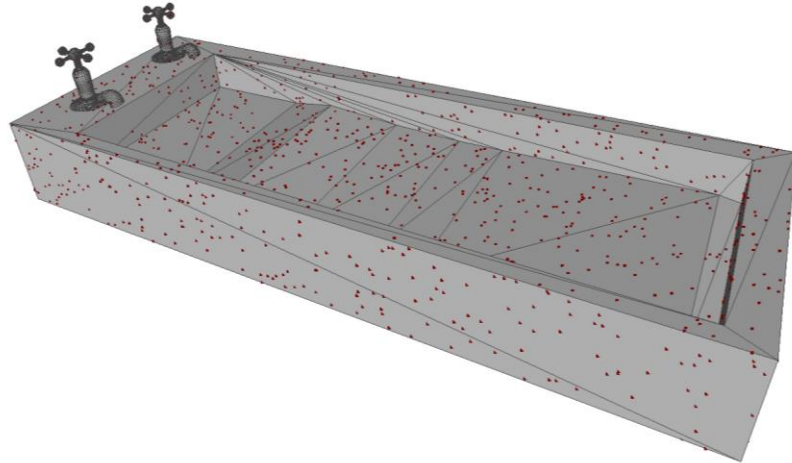
Example for bathtub:

Sample from Class bathtub (misclassified)

Top 1 prediction: bed (52.32%)

Top 2 prediction: bathtub (24.64%)

Top 3 prediction: sofa (21.42%)



Example for nightstand:

Sample from Class night_stand (misclassified)

Top 1 prediction: dresser (64.87%)

Top 2 prediction: night_stand (33.99%)

Top 3 prediction: monitor (0.64%)

